

AVINASH K. RAVISHANKAR

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EDUCATION:

Rice University (Houston, TX) - M.S. Data Science

Expected December 2026

Coursework: Machine Learning, Computer Vision, Deep Learning, Bayesian Methods, Big Data (PySpark/AWS EMR), DevOps Tools

Teaching Assistant: COMP 665 Data Visualization — Python (matplotlib, Plotly, seaborn), Tableau, Power BI, time series analysis

Data Competitions: Hex Data Hackathon | NCL Cybersecurity CTF (log forensics, network analysis, data challenges)

Rice University (Houston, TX) - B.S. Mechanical Engineering

2008-2012

Activities & Coursework: Society of Automotive Engineers (SAE), Rice Robotics Team, MATLAB, System Design

INTERN EXPERIENCE:

Computer Vision Research Intern | Digital Health Institute & Rice University (Houston, TX)

May 2026 - Present

Tissue material characterization for autonomous surgical robots

- Developed 500 GPU-accelerated soft-body physics simulations using the Genesis simulator in Python to model tissue deformation and material properties relevant to autonomous robotic surgery
- Studied and implemented 5 cutting-edge research papers including NVIDIA's DiSECT, GIC (NeurIPS 2024), and MASIV to understand differentiable simulation techniques for estimating material properties from video
- Exploring machine learning pipeline to identify tissue material characteristics (stiffness, elasticity) from RGB video using physics-based simulation, with applications in surgical robotics and haptic feedback systems

Data Science Intern | Houston Texans (Houston, TX)

September 2025 - October 2025

4 week data science project to define Explosive Plays sponsored by Strength in Numbers

- Built Python (scikit-learn, pandas, numpy) classification models achieving 82% accuracy on 600+ sequential events using time-series features from spatial positioning and velocity data
- Developed automated SQL pipeline for sensor measurement extraction, presented recommendations to operations team

WORK EXPERIENCE:

Sr. Product Manager | TaskTag (Houston, TX)

Jul 2022 — May 2025

Mobile and Web app for construction project management through chat

- Managed distributed engineering team (12 engineers, 2 QA across 8 countries) to launch mobile/web application acquiring 2,000+ users within 3 months; built SQL analytics dashboards and conducted A/B testing to optimize product features

VP of Product | Mesos (Houston, TX)

Mar 2021 — Aug 2022

Autonomous stratospheric drone and aerial imagery with computer vision and machine learning

- Led 3-person team designing stratospheric UAVs with autonomous imaging systems integrating camera, GPS, and barometric altitude sensors for real-time navigation and asset tracking
- Developed computer vision algorithms using CNNs in Python (TensorFlow) for automated object detection from aerial platforms, saving customers average \$10,000 annually

Sr. Product Manager | daytum (Houston, TX)

Jan 2019 — Feb 2021

Online data science education. Software to teach data science and machine learning to energy professionals

- Built Learning Management System in Anaconda (Jupyterhub, Postgres, Docker) serving 500+ engineers across 12 energy companies. Designed curriculum alongside technical team. Topics ranged from linear regression to neural networks and Pytorch. Recorded and distributed 50+ hours of data science education content.

Head of Product & Cofounder | Muvve (Houston, TX)

May 2016 — Jan 2019

Fitness events marketplace. B2B2C Product. Connecting fitness enthusiasts to local fitness events

- Managed 2 engineers and 1 designer to launch iOS marketplace with 1,000+ downloads
- Analyzed user behavior patterns from 20,000+ contacts using SQL
- Optimized feature development and drove 10% revenue increase for 20+ partner companies using product analytics

Mechanical Engineer | OXY Occidental Oil and Gas

Jun 2009 – Aug 2015

Oil and Gas E&P Upstream (Permian Basin, Bogota, Bakersfield)

- Applied risk modeling software and sensor data analysis across 200+ miles of pipeline infrastructure and 50 well sites; managed large-scale corrosion datasets for predictive maintenance and failure prevention